

UIDAI HACKATHON

Aadhaar Migration Intelligence: Understanding India's Internal Movement Patterns

TEAM:

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Github code link - <https://github.com/rachana192837/UIDAI-Hackathon.git>

PROBLEM STATEMENT AND APPROACH

India sees massive internal migration every year—people move for jobs, education, climate stress, and family reasons.

Here's the thing: when people migrate, many update their address in the Aadhaar system to access services in their new location. These routine updates, when looked at in aggregate form might actually tell us where people are moving in near-real time.

Core Problem: Can Aadhaar address and demographic updates be used as a near-real-time signal of internal migration and shifting service demand across regions?

APPROACH

This project adopts a data-driven analytical approach using the anonymised Aadhaar enrolment and demographic update datasets provided by UIDAI. Address-related demographic updates were used as a proxy indicator for internal migration and were cleaned, filtered, and aggregated temporally (monthly) and spatially (state/district) to construct migration timelines and regional intensity measures.

- Additional features such as mobile number updates, child enrolments, service-type distributions, and seasonal migration patterns were engineered to enable univariate, bivariate, and multivariate analysis. Correlation analysis and statistical tests were applied to examine relationships between migration intensity, digital inclusion, and family mobility.
- we developed an univariate ARIMA time-series model to forecast short-term migration trends.
- Finally, we did an interactive executive dashboard to consolidate trends, regional hotspots, seasonal effects, equity distribution, and forecasts into a single decision-support interface for policymakers.

2. DATASETS USED

We used the anonymised Aadhaar datasets provided by UIDAI:

2.1 Aadhaar Enrolment Dataset

Key Columns Used:

- enrolment_date
- state / district
- age_group
- gender
- enrolment_type

2.2 Aadhaar Demographic Update Dataset

Key Columns Used:

- update_date
- update_type (address, mobile, name, DOB)
- state / district
- age_group

2.3 Biometric Dataset

- month (from date)
- migration_flag (1 if address update)
- equity_category (low/medium/high service access)

3. METHODOLOGY

3.1 Data Ingestion and Cleaning

The anonymised Aadhaar enrolment and demographic update datasets provided by UIDAI were loaded into Python using the Pandas library. All date fields were converted into a standard datetime format to enable reliable time-based analysis. Records containing missing, invalid, or inconsistent date values and geographic identifiers (state or district) were removed to maintain temporal and spatial accuracy.

State and district names were cleaned and normalised to handle spelling variations and formatting inconsistencies that could otherwise lead to duplicate or fragmented regional records. Finally, only the columns relevant to migration analysis and related socioeconomic indicators were retained, ensuring a clean and focused dataset for subsequent analysis.

3.2 Construction of Migration Proxy

We consider the

The address-related demographic updates were considered as indicators for internal migration events. The records which contain address change indicators were extracted from the dataset for demographic updates.

These address update records were aggregated:

- Temporally (by month): to produce a national migration pattern or timeline.
- Spatially (State/District level) to calculate the intensity of migration.
- This process enabled us to take unprocessed Aadhar update log data and distill it into organized migration indicators

3.3 Feature Engineering

Multiple derived features were created to support deeper analysis:

- Monthly migration volume
- Regional migration intensity
- Mobile number update counts
- Child enrolment update counts
- Seasonal migration indices
- Migration growth rates
- Service-type distributions
- Composite equity index (based on migration and service-access intensity)

These capabilities allowed for both cross-sectional, bivariate, and multivariate analysis into various time periods, geographic, and service distributions.

3.4 Socioeconomic Impact & Correlation Analysis

To quantify socioeconomic effects of migration:

Pearson correlation coefficients were calculated for:

- Migration volume and mobile number updates
- Volume of migration & Enrollment of Children

Comparisons on a regional level have been used to examine the

- High Migration Rates but Low Access
- Car-sharing schemes
- Excluded transport clusters

Such analyses were employed to research the relationship between migration, digital inclusion, and family mobility.

3.5 Seasonal Pattern and Anomaly Analysis

Monthly aggregation enabled the detection of:

- Seasonal spikes in address updates
- Periodic migration surges
- Region-specific volatility

These patterns were interpreted as indicators of seasonal labour migration and potential distress-migration events.

```
# By month (seasonal patterns)
migration_monthly = migration_data.groupby('month').size().reset_index(name='migrations')
migration_monthly['month_name'] = migration_monthly['month'].map({
    1: 'Jan', 2: 'Feb', 3: 'Mar', 4: 'Apr', 5: 'May', 6: 'Jun',
    7: 'Jul', 8: 'Aug', 9: 'Sep', 10: 'Oct', 11: 'Nov', 12: 'Dec'
})
```

3.6 Time-Series Forecasting

A univariate ARIMA time series model was used and trained on the array of data that was used to plot the historical migration proxy timeline, so as to provide a prediction regarding the migration intensity that would be experienced in the future. The model was employed to:

- Determine expected trends regarding migration growth or stability
- Support forward-looking service planning and policy interventions

```
try:
    # Simple ARIMA model
    model = ARIMA(ts_data_smooth, order=(2,1,2))
    model_fit = model.fit()
```

3.7 Executive Dashboard Development

An interactive executive summary dashboard was created using the Plotly tool.

It integrates:

Migration Schedules

Temperature hotspots

Seasonal effects

Socioeconomic Cor

Equity Index Funds

Forecasted migration trends

Statistical significance tests

The dashboard can be used as a decision support instrument for policymakers and administrators.

4. DATA ANALYSIS AND VISUALISATION

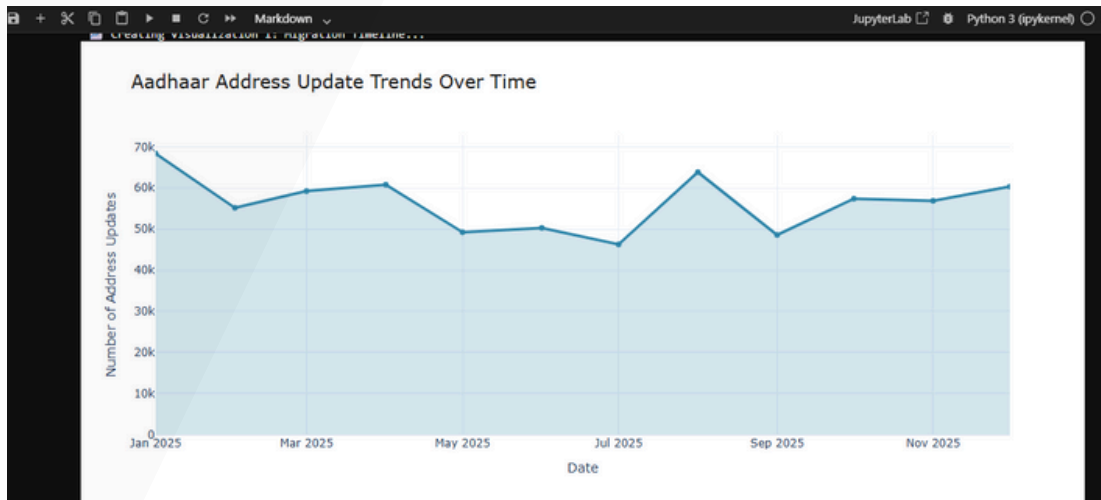
4.1 Migration Timeline Analysis

A national monthly migrate proxy timeline has been created using address-related Aadhaar updates.

Key Observations:

Updates for human addresses show strong time variation with multiple distinct spikes.

- An increasing trend shows that there are more people on the move and, consequently, a growing demand for services.

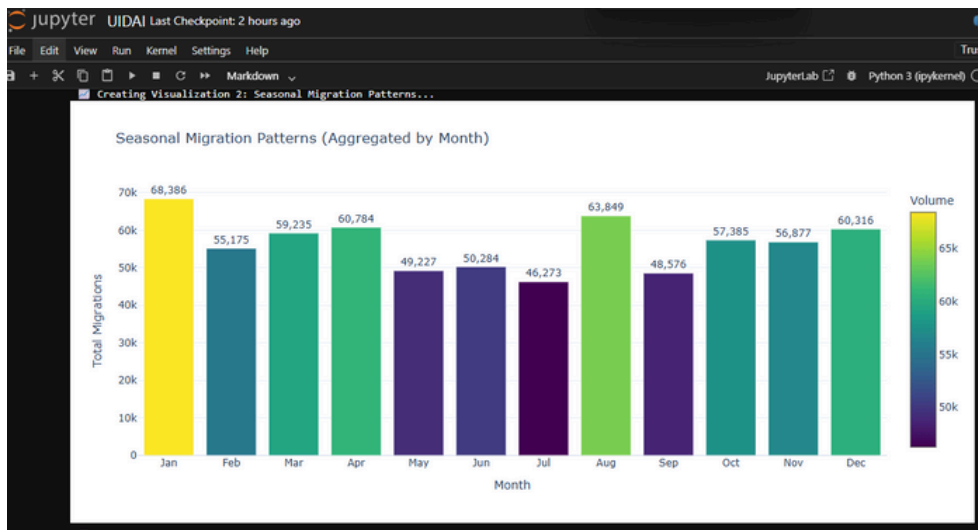


4.2 Seasonal Migration Patterns

The aggregation by month over the years showed steady seasonal cycles.

Key Observations:

- The magnitude of migration is at its peak in particular months every year.
- “These peaks are almost certainly associated with the cycles of agricultural activity, the educational cycles, and the cycles of labor migration.”
- The effects of seasonality were more pronounced in states where migration was higher

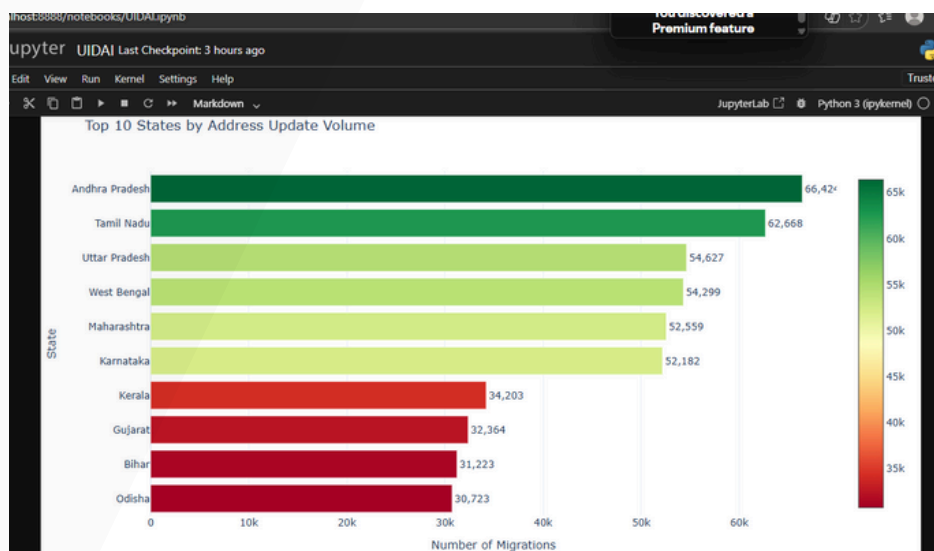


4.3 Regional Migration Hotspots

Aggregation by state and district identified regions of high

Key Observations:

- A few states & their corresponding districts are responsible for most migration-related updates.
- Urban and industrial areas have a high rate of migration.
- A number of regions of rural origins show continuous patterns of out-migration

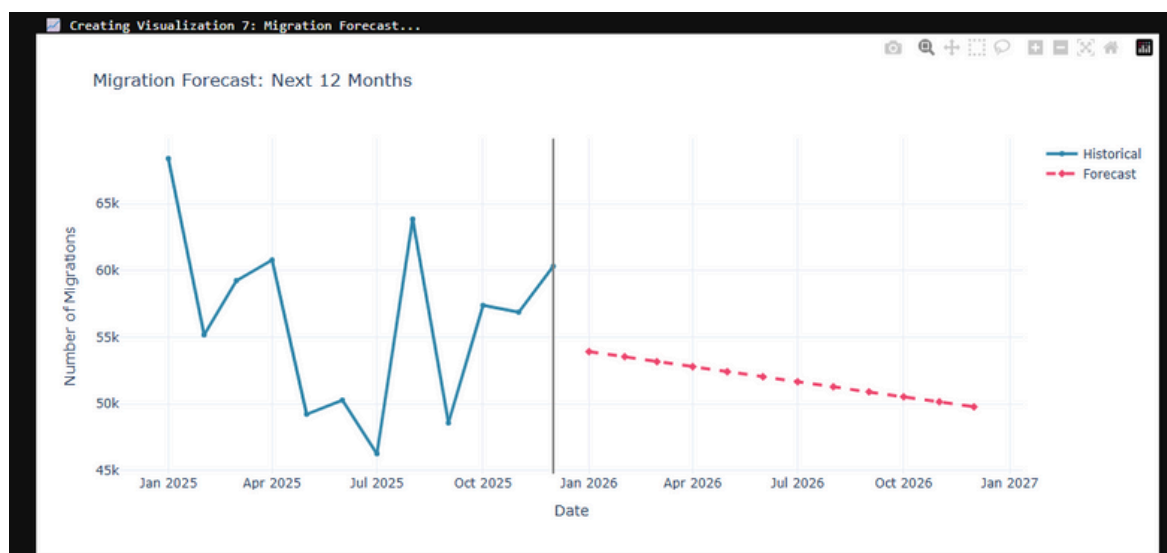


4.4 Forecasting Future Migration

The trends of migration proxies were forecasted by using an ARIMA model.

Important Points:

- "Forecasts also show sustained migratory pressures in urban and industrial states."
- Migration flow will not see a sharp drop in strength in the short-term period.
- This points towards a continuous requirement for Aadhaar updation services.



Executive Summary Dashboard

A consolidated executive dashboard was developed to summarize all major findings.

Dashboard Components:

- Migration timeline
- Regional hotspots
- Seasonal patterns
- Child enrolment impact
- Equity distribution
- Statistical significance
- Migration forecast



JUPYTER NOTEBOOK (GITHUB LINK)

The complete Jupyter notebook containing all data preprocessing, analysis, forecasting, and visualisation code used in this project is available at:

